ENVIRONMENTAL HACKS | BHARAT BUILDS TOUR 2026 | TRACK 03: WASTE AND ENERGY

SolarSabha

The AI project manager for your society's solar roof.

Research and Learning Guide

Everything about the idea in plain language: the real problem, the research behind it, how AI agents actually work (explained step by step), the full tech stack, and a four-day plan. Written so that anyone on the team can follow it.

Team: [Team name]



VERSION	DATE	BUILD DATES	TRACK
Draft v2	4 Oct 2026	8-11 Oct 2026	03 | Waste and Energy

START HERE

1 The whole idea in one minute

Many apartment societies pay a big electricity bill every month just to run **lifts, water pumps, corridor lights and gates**. A solar system on the society's terrace can cut that bill by a lot, and the government pays part of the cost. So why does almost every society not do it?

Because it is not a buying decision. It is a **months-long project** with many people who must agree and many offices that must reply: the committee, 50 to 500 flat owners, solar vendors, the electricity company (called the **DISCOM**), and a government portal. Papers get lost, a vendor says one thing and the rules say another, and nobody chases the DISCOM when it goes quiet. Most projects die of tiredness, not of bad maths.

In one sentence
SolarSabha is an AI **agent** that acts like a patient, well-informed project manager for the society's committee. It reads the electricity bills, works out the right solar size, writes the papers for the general body meeting (the **sabha**), checks vendor quotes for traps, and then keeps following up for weeks until the subsidy lands. People stay in charge of every decision. The agent does the reading, calculating, drafting and nagging.

How to use this guide

- **Sections 2 to 4** explain the problem and the research. No technical words.
- **Sections 5 to 9** follow a real user, then show what SolarSabha does, a worked example you can check by hand, the quote audit and the negotiation arena.
- **Sections 10 to 18** are the learning part: what an AI agent really is, how it works, how our agent team and MCP server are built, how agents wait for weeks, and how to make them trustworthy. Read these slowly. They are the skill you take away from this hackathon.
- **The Tech Stack page** comes next, followed by Sections 19 to 25: the four-day plan, the video script, an honest expert review, risks, the **repository guide** for the team, a glossary and sources.

A note on trust
Facts in this guide come from news reports, government scheme documents and company guides, all listed at the end. Numbers marked *illustrative* are our own examples. Rules and amounts differ by state and can change, so SolarSabha is designed to quote the rule it used and ask a human to confirm. We found that vendor blogs sometimes disagree with each other, and we say so where it matters.

2 What is society solar, in plain words?

Imagine a building with 50 flats. Each flat has its own meter and its own bill. But the building also has a **common meter** that pays for everything shared: lifts, water pumps, staircase lights, gate, garden lights, and maybe EV chargers. That bill is paid from the society's maintenance fund, which means every family pays a share of it every month.

A solar system on the terrace makes electricity in the daytime, exactly when the pumps and lifts are working. Whatever the panels make, the society does not have to buy from the grid. Less bill means lower maintenance charges, or more money for repairs and gardens.

A real-world example
A housing society that reported paying about Rs 18,000 a month for common-area electricity said the bill fell to about Rs 3,000 a month after solar. This is one society's own account in a news feature, so treat it as an example, not a promise.

What does the government give?

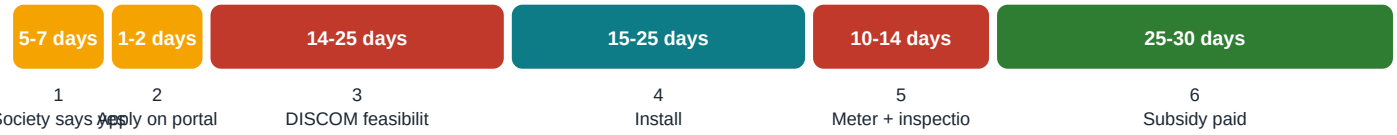
Under the national scheme **PM Surya Ghar: Muft Bijli Yojana**, a group housing society or resident welfare association (RWA) can get **Rs 18,000 for every kilowatt (kW)** of solar it installs for common facilities, including EV charging, up to 500 kW. The 500 kW limit is worked out at 3 kW per flat, and it also counts any solar that individual flat owners have already installed. (A kilowatt is just a measure of the size of the system. A 10 kW system has roughly 20 to 25 panels.)

WHAT	NUMBER	SOURCE
Subsidy for society common facilities	Rs 18,000 per kW, up to 500 kW (3 kW per flat)	Official scheme subsidy sheet (Sources 1)
Societies already paid out	19,035 societies, Rs 422.30 crore, 9.09 lakh households	News report, 12 days before this guide (Sources 2)
Typical electricity made	About 1,400 to 1,800 units (kWh) per kW per year	Company guides (Sources 6, 7)
Typical installed cost	About Rs 45,000 to Rs 60,000 per kW	Company guides (Sources 6, 7, 8)
Grid pollution factor	0.710 tonnes of CO2 per MWh (FY 2024-25)	Central Electricity Authority (Sources 5)

Important correction
The government cap (3 kW per flat) is only a **ceiling**. The system should be sized to what the common meter actually uses, because the subsidy is for common facilities. A 60-flat society has a ceiling of 180 kW, but that does not mean it should buy 180 kW. We got this wrong in an earlier chat note, and SolarSabha's sizing step is built to avoid exactly this mistake.

3 Why a good idea still gets stuck

Here is the journey every society must walk. Look at the red parts: they are where days disappear.



Red = the stages where waiting and chasing usually happen. Day ranges come from vendor guides, not the government.

STAGE	WHAT HAPPENS	WHO IS INVOLVED
1 Society says yes	The committee calls a general body meeting (AGM). Residents vote. The society needs a resolution, a cost-sharing method, and where panels will go.	Committee, all flat owners
2 Apply on portal	The society registers on the national rooftop solar portal with bills, bank details and documents.	Secretary, vendor
3 DISCOM feasibility	The electricity company checks the roof, the sanctioned load and the connection, then approves.	DISCOM
4 Install	An approved installer fits the panels.	Vendor
5 Meter and inspection	The DISCOM fits a net meter, inspects, and issues a commissioning certificate.	DISCOM, vendor
6 Subsidy	After commissioning, the subsidy is sent to the society's bank account.	Portal, bank

What the research says goes wrong

- **Delays.** The scheme promises a 30-day window for approval and commissioning, but an energy think tank found real timelines often run 45 to 120 days, with causes like meter shortages and poor coordination between vendors and DISCOMs (Sources 3). One guide says the 30-working-day promise covers only the final payment stage after commissioning.
- **Missed appointments.** One guide says three missed DISCOM inspection appointments can lead most DISCOMs to cancel the application (Sources 9). A person who is also running a job and a family easily misses one.
- **Paperwork traps.** Guides warn that a non-empanelled installer can void the subsidy, that signing with an installer before DISCOM approval can hurt eligibility, and that a bank account not linked to Aadhaar is a common reason for payment delays (Sources 9, 10). These come from commercial guides, so SolarSabha treats them as warnings to verify, not final law.
- **The group problem.** A recent ground report points out that a shared terrace needs a collective decision: how it is paid for, where panels go, how electricity is used, and how savings are shared (Sources 4).
- **Thin vendor experience.** In one Ghaziabad neighbourhood, four vendors said they had not handled a single society installation under the scheme (Sources 4). That is one area, not all of India, but it shows society projects are not routine for everyone.

The key insight
Nobody is doing something wrong. The project simply has too many moving parts, and the person responsible, the society secretary, is a volunteer. What the project needs is someone who **reads everything, remembers everything, double-checks, and follows up without getting tired**. That is a job description for an agent.

4 What already exists, and the gap we are filling

We should be honest about this, because judges will ask. Here is what we found:

WHAT EXISTS	WHAT IT DOES	WHAT IT DOES NOT DO
Vendor and company websites	Explain the scheme. Some offer a free report or calculator for a society.	Nobody follows the project for you. Their goal is to sell a system.
Solar calculators	Give size, cost and payback from a monthly bill.	Do not read your actual documents, audit quotes, write the AGM papers or track stages.
The national portal	Where the application is filed and tracked.	Does not explain the next step in plain words or chase the DISCOM for you.
Consultants	Do the whole job, for a fee.	Hard for a small society to afford or to trust, and still not automatic.

The gap
SolarSabha is not a calculator and not a seller. It is **a neutral project manager for the society's committee**. Its job is to make the committee well informed, hard to mislead, and organised. We did not find a tool that runs the whole society process end to end, but our search was limited to what is easy to find online. Before pitching, ask mentors and look again.

Why this fits the hackathon

WHAT JUDGES ASK	OUR ANSWER
Does it fix a real environmental problem?	Yes. More rooftop solar replaces coal-heavy grid electricity. We measure kW installed and CO2 avoided using the official CEA factor.
What changes for the people who live with it?	Lower maintenance bills, fewer months lost, fewer costly mistakes.
A small problem solved well	One type of project (society common-area solar), one state's rules, one clear loop.
Would someone who is not on our team know what to do?	The committee uploads bills. The agent asks plain questions. Everything is explained in simple words.
Built on AWS	Bedrock, Strands Agents SDK, Lambda, Step Functions, EventBridge, DynamoDB, S3, SES. See the Tech Stack page.

PART B | THE IDEA

5 Meet Meera: the real user

Meera, 41, is the volunteer secretary of a 50-flat society. She has a full-time job, a family, and has never dealt with a DISCOM. She does not want to 'use an agent'. She wants solar approved without being cheated and without losing six months. (Meera is an invented example. One real conversation with a society secretary is worth more than any slide, and we should get one before the video.)

WHEN	WHAT MEERA DOES	WHAT SOLARSABHA DOES	REAL OR SIMULATED IN THE DEMO
Week 1	Photographs 12 common-area bills and the terrace. Uploads them.	Reader reads the bills and flags any unclear month. Code sizes the system and shows cost, subsidy, payback and CO2.	Reading and maths are real.
Week 2	Asks 3 or 4 vendors for quotes.	Drafts the request-for-quote emails. Meera approves and sends them herself.	Drafts are real. Nothing is sent by the app.
Weeks 2 to 3	Receives quote PDFs by email.	Reader extracts them. Auditor flags risks. The agent drafts counter-offers for Meera to approve.	Reading and flags are real. Vendor replies are simulated.
Week 3	Calls the general meeting (AGM).	Meeting notice, resolution text, and a table of what each flat pays and saves under different sharing rules.	Real.
Weeks 4 to 10	Waits for the DISCOM, the part that kills projects.	Chaser tracks the stages, drafts a reminder, then an escalation when the delay grows, and re-plans if the DISCOM limits capacity.	The clock and the DISCOM's reply are simulated.

What problem is actually solved
The friction that kills these projects: not knowing the numbers, not spotting a bad quote, running a group decision, and chasing for weeks. SolarSabha removes that friction. It does **not** install panels and it cannot make a DISCOM faster. It makes the committee informed, hard to mislead, and organised.

Why parts of the demo are simulated
Real vendors do not talk to an AI agent live. Real negotiation is days of email. So in the demo the vendor side is simulated and labelled, and everything that happens on the committee's side (reading, auditing, drafting, tracking) is real.

6 What SolarSabha does

Think of five jobs that a good project manager would do. SolarSabha does the same five, one after another.



JOB	IN PLAIN WORDS	WHAT THE COMMITTEE SEES
1 Understand	The committee uploads the last 12 common-area bills (photos or PDFs) and a rough roof size. The agent reads the numbers out of them and lists anything unclear.	A clean table of units and bill amounts, with 'I was not sure about this' flags.
2 Size	It works out how many kW fit the society's actual use and roof, how much the subsidy is, what the society pays, how fast it pays back, and the CO2 saved. Plain code does the maths, not the AI.	One page with three options (small, recommended, large) and why.
3 Decide	It drafts the notice for the general body meeting, the resolution wording, a cost-sharing proposal, and a no-objection letter, ready for the secretary to edit.	Documents to download and edit.
4 Choose	The committee uploads vendor quotes. The agent compares them on a like-for-like basis and flags risks: price per kW far from the norm, vendor not on the approved list, big advance payment, pressure to sign before approval.	A side-by-side table with red, amber and green flags and the reason for each.
5 Follow	After the vote, it builds the timeline, sets reminders, and drafts the messages: a nudge to the vendor, a polite reminder to the DISCOM, an escalation to the grievance cell when a deadline is missed.	A timeline board with 'next step', 'who is late' and a draft message ready to send.

Human in charge
The agent may read, calculate, compare, draft and remind. **The agent may not** sign anything, pay anyone, submit on the government portal for you, or send an email to an official without a human clicking approve. This is a design choice, not just a safety rule: it keeps the society in control and keeps us honest.

7 A worked example you can check with a calculator

This example is **illustrative**. Every assumption is listed, so you can change it and see how the answer moves. This is also the kind of output SolarSabha produces for a real society.

WHAT WE ASSUME	VALUE	WHERE IT COMES FROM
Flats in the society	50	Example
Common-area electricity used	5,000 units (kWh) a month, so 60,000 a year	Example (from the bills)
Electricity made per kW per year	1,400 units	Low end of the 1,400 to 1,800 range (Sources 6, 7)
Share of the bill we aim to cover	85%	Sensible target, so we do not oversize
Installed cost	Rs 50,000 per kW	Middle of the Rs 45,000 to 60,000 range
Subsidy	Rs 18,000 per kW	Official subsidy sheet (Sources 1)
Price paid for grid electricity	Rs 7 a unit	Example; varies by state and tariff class

Step by step

STEP	SUM	RESULT
Units we want solar to cover	60,000 x 85%	51,000 units a year
System size needed	51,000 / 1,400	about 36.4 kW, so we choose **36 kW**
Check the subsidy ceiling	50 flats x 3 kW	150 kW allowed, and 36 kW is below it. Good.
Electricity made	36 kW x 1,400	50,400 units a year
Total cost	36 kW x Rs 50,000	Rs 18.00 lakh
Subsidy	36 kW x Rs 18,000	Rs 6.48 lakh
Society pays	cost minus subsidy	**Rs 11.52 lakh**
Yearly saving	50,400 units x Rs 7	Rs 3.53 lakh a year
Simple payback	11.52 / 3.53	about **3.3 years**
CO2 avoided	50.4 MWh x 0.710	about **36 tonnes a year**

Reality check
This is a simple payback. It ignores maintenance, panels slowly losing output, and fixed charges. Company guides quote 3 to 6 years for society projects, so our figure is on the optimistic side. SolarSabha should always show a range and say which assumptions it used.

8 The quote audit: where the agent earns its keep

Imagine the society collects three quotes. Here is the kind of thing the agent can spot that a tired volunteer might miss. (These quotes are invented for the demo.)

	VENDOR A	VENDOR B	VENDOR C
Price for 36 kW	Rs 17.5 lakh	Rs 14.0 lakh	Rs 19.0 lakh
Price per kW	Rs 48,600	Rs 38,900	Rs 52,800
On the approved vendor list?	Yes	**Not found**	Yes
Payment terms	10% advance, rest at stages	20% advance	**50% advance before approval, 'sign today' offer**
Warranty and upkeep	25 years panels, 5 years service	Not stated	25 years panels, 1 year service
Agent's verdict	**GREEN**: fair and clear	**RED**: cheapest, not on the list, and its file hides a command aimed at the AI	**AMBER**: dearest, with pressure tactics

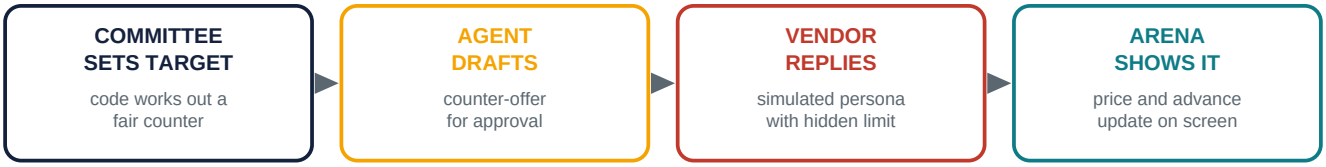
How the agent reaches a verdict

- It reads each PDF and pulls out the same list of fields every time (price, capacity, brand, payment terms, warranty).
- Plain code calculates price per kW and compares it with a sensible band. The AI is **not** trusted to do the arithmetic.
- It looks up the vendor against an approved-vendor list we load into the system, and it says plainly when it **cannot verify** something.
- It checks payment terms against rules like 'no large advance before approval', quoting the rule and where it came from.
- It writes a short, human explanation for each flag and suggests a question to ask the vendor.

Honest limit
For the demo we will load a small approved-vendor list that we prepared ourselves and label it clearly as sample data. In real use, the official list comes from the portal. Never let the agent claim that a vendor is approved unless it found it in the list.

9 The negotiation arena (simulated, animated)

After the audit, the agent helps the committee negotiate. In the app this is shown as an animated scene: the committee at a table, three vendors around it, and offers that fly across the table round by round, with each vendor's current price shown above its seat.



RULE OF THE ARENA	WHY
A vendor that failed verification (RED) is not negotiated with. The agent asks for proof of empanelment first.	Haggling with an unverified installer helps nobody and could mislead the committee.
The agent only uses quotes the committee actually uploaded. It never invents a competing quote.	Invented leverage would be dishonest and would fall apart in real life.
Every counter-offer is a draft. A person approves it.	The committee, not the agent, makes commitments.
The scene is labelled SIMULATION and the vendors are personas with hidden floors.	Real negotiation takes days by email. The scene is a replay of the logic, not a claim of savings.

Why animated 2D and not full 3D

Judges reward a clear story, not graphics. A true 3D scene can easily cost a whole day. A 2.5D scene (a table drawn with perspective, avatars, flying offer bubbles) gives most of the effect for a fraction of the work. In the repository it is plain SVG and CSS, driven by the same events the agents produce. If time allows, a polished 3D version can be added later.

Honesty point
Do not claim 'the agent saved the society Rs X lakh' in the video. Say: 'the agent drafts counters and shows how a negotiation could unfold'. The honest version is still impressive and cannot be punctured by a judge.

10 Chatbot, workflow, agent: what is the difference?

You will hear 'agent' used for everything. Here is a simple way to tell them apart, using a restaurant as an example.

	CHATBOT	WORKFLOW	AGENT
Like a...	Waiter who answers questions about the menu	Kitchen conveyor belt: same steps, same order, every time	Restaurant manager: gets a goal, decides what to do, uses staff and tools, checks the result
Who decides the steps?	You, one message at a time	The programmer, in advance	The AI, while working, within rules
Can it use tools?	Usually no	Yes, but fixed	Yes, and chooses which and when
Good for	Answering and explaining	Predictable jobs	Messy jobs with changing situations
Risk	Wrong answer	Breaks when something unexpected happens	Can wander or act wrongly if not controlled

Our working definition
An **AI agent** is an AI model that can **decide what to do next**, use **tools** to do it (read a file, run a calculation, look something up, draft a message), **look at what happened**, and keep going until the goal is met or it needs a human.

Is SolarSabha really an agent, or just a workflow?

A fair question, and judges may ask it. Parts of SolarSabha are plain workflows (read bill, calculate size). It becomes agentic where the situation is **not predictable**: a bill is missing a month so the agent must ask, a quote has an unusual clause so it must investigate, the DISCOM goes quiet so it must decide when and how to escalate, a vendor replies with something new so the plan must change. A good design uses fixed steps where the job is fixed, and an agent where judgement is needed.

11 The agent loop: think, act, look, repeat

Almost every agent, whatever the framework, runs the same loop. Understanding this loop is the single most useful thing in this guide.



A tiny example of the loop in SolarSabha:

ROUND	THE AGENT THINKS	THE AGENT DOES	WHAT IT SEES
1	I need the numbers from Quote B first.	Calls tool: read_quote(B)	Price, 20% advance, warranty missing
2	Is price per kW normal?	Calls tool: price_per_kw(B)	Rs 38,900, well below the others
3	Is this vendor on the approved list?	Calls tool: check_vendor('B')	Not found
4	Cheap, unverified, and no warranty. I should warn the committee.	Writes the red flag and a question for the vendor	Done. Shows result to human.

12 The six parts of an agent

Every agent is made from the same six parts. When something goes wrong, the problem is almost always in one of them.

PART	WHAT IT IS (ANALOGY)	IN SOLARSABHA
1 The brain (the model)	The thinker. A large language model that reads and writes text and decides what to do.	A Claude model on Amazon Bedrock.
2 Instructions (the system prompt)	The job description and rules you hand to a new employee on day one.	'You help a housing society committee. Never claim a vendor is approved unless found in the list. Always cite the rule. Never send anything.'
3 Tools (the hands)	Things the agent can actually do: functions in code. Without tools an agent can only talk.	read_bill, size_system, price_per_kw, check_vendor, draft_notice, set_reminder.
4 Memory and state (the notebook)	What the agent remembers about this project, saved so it survives days and restarts.	A project record in DynamoDB: bills, chosen size, quotes, stage, dates.
5 Guardrails (the fences)	Hard limits that hold even if the AI is confused.	Draft-only email tool. No payment tool exists. Maths done in code.
6 Human checkpoints (the manager)	Points where the agent must stop and ask a person to approve.	Before sending any message to a vendor or official; before finalising a recommendation.

The most important design lesson
Rule of thumb for builders: let the AI do what only an AI can do well (read messy documents, write clear text, decide what is worth flagging) and let ordinary code do everything that must be exactly right (arithmetic, rules, dates, permissions). Many weak agent projects fail by asking the AI to do both.

13 Tools: what they look like in code

A tool is just a normal Python function with a clear description. The agent reads the description to decide when to use it. This is a short, **illustrative** sketch using the Strands Agents SDK, which is AWS's open-source agent framework. Its documentation shows agents built from an Agent object, a tools list and a model, with tools declared using a decorator (Sources 11).

```
from strands import Agent, tool
from strands.models import BedrockModel

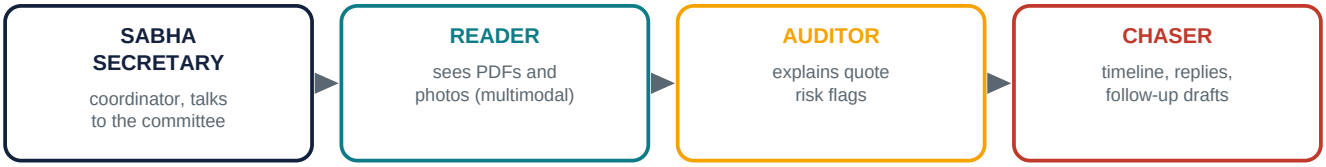
@tool
def size_system(annual_kwh: float, flats: int) -> dict:
    """Work out a sensible solar size in kW for common-area use."""
    kw = round(annual_kwh * 0.85 / 1400)      # plain code, not AI
    ceiling = flats * 3                         # scheme ceiling
    return {"kw": min(kw, ceiling), "ceiling": ceiling}

agent = Agent(
    model=BedrockModel(model_id="<a model enabled in your account>"),
    tools=[size_system],
    system_prompt="You help a housing society plan solar. Cite rules. Never send anything.",
)
agent("We use 60000 kWh a year and have 50 flats. What size?")
```

Notice the three ideas: the maths lives in the tool; the description (the docstring) tells the agent when to use it; and the system prompt sets the rules. That is a complete tiny agent.

14 SolarSabha's team of agents

An earlier draft of this guide had five specialist agents. An honest review (Section 21) showed that was too many: one of them did plain maths that does not need an AI at all. The final design is a coordinator plus **three** specialists, backed by plain-code tools.



AGENT	WHAT IT SEES OR HEARS	ITS JOB	TOOLS	MAY IT ACT ALONE?
Sabha Secretary	The committee's requests (text)	Decides which specialist to call, summarises, asks for approvals.	Reader, Auditor, Chaser (as tools), size_system, money_summary, get_rule	Reads and drafts only.
Reader	Photos and PDFs of bills and quotes (multimodal)	Turns documents into labelled data. Marks anything unclear. Never guesses.	None. Returns a fixed form.	Yes (read-only).
Auditor	Extracted quotes	Explains the risk flags in plain words and suggests a question to ask the vendor.	audit_quote, check_vendor, get_rule	Advice only.
Chaser	DISCOM replies, timeline	Tracks stages, classifies replies, drafts reminders and escalations, re-plans.	build_timeline, next_action, classify_reply, replan, draft tools	Drafts only.

The plain-code tools (these are not agents, on purpose)

TOOL	WHAT IT DOES	WHY IT IS CODE
size_system, money_summary	Size, cost, subsidy, payback, CO2	Must be exactly right and testable.
fairness_table	What each flat pays under equal or by-area sharing	Arithmetic.
audit_quote	Price per kW, vendor check, payment and warranty flags, hidden-instruction scan	Rules should be the same every time.
build_timeline, chase_level	Which stage is late and how hard to chase	Dates and thresholds.
run_simulated_negotiation	Counter-offer rounds for the arena	Deterministic, so the demo is repeatable.
drafts	Fills letter templates with facts	Facts must not change. The AI may polish wording only.

Why a coordinator with three specialists, not seven agents

- **Fewer moving parts.** Four agents are easier to test, debug and explain in a 3-minute video.
- **Right tool for the job.** Anything that must be exact is code. Agents do reading, explaining, deciding and drafting.
- **Safer.** Only the Chaser touches communication, and it can only draft.
- **Pattern:** the Secretary calls specialists as tools ('agents as tools' in Strands). Other patterns exist (swarm, graph, workflow) but are not needed here.

Roadmap, not this week
Possible later additions, not for this week: a Terrace Scout that reads rooftop photos (risky: a wrong usable-area estimate is worse than none), and a Voice agent for Hindi or English voice notes. Say they are on the roadmap, do not build them now.

15 MCP: plugging tools into agents

MCP (Model Context Protocol) is a standard way for an AI agent to use outside tools, a bit like a universal plug socket. Instead of writing special code for every tool, a tool is put behind an MCP server once, and any agent that speaks MCP can use it.

Where we use it, and where we do not

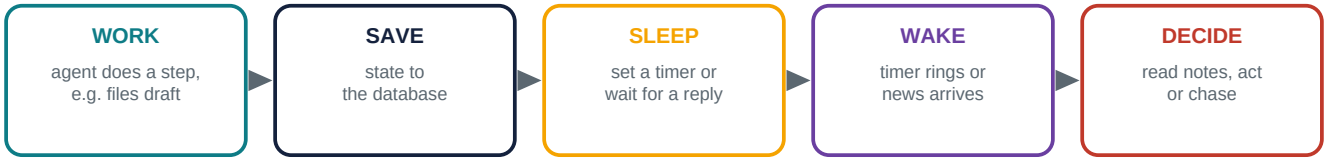
IDEA	REAL USE?	OUR DECISION
SolarRules MCP server: our own small server that exposes the rulebook, the calculators, the vendor check and the hidden-instruction scan as MCP tools	Yes. The same trusted rules and maths are available to any agent, including one built by another team.	**Build it.** It already exists in the repository and is tested.
Gmail or Calendar connectors to draft real emails and set the AGM date	Useful in a real product, but sign-in setup can eat hours.	Only if the core works by Saturday. Drafts only, never send.
Google Drive to save the generated documents	Nice, but not essential.	Skip for the hackathon.
MCP on every part just to say we used it	No.	Do not do this. Judges see through it.

In the code
The Strands Agents SDK can connect to an MCP server and load its tools straight into an agent. In the repository, `agents/mcp_tools.py` does this with a short context manager, and a test starts the server and calls a tool over MCP to prove it works.

16 How can an agent wait three weeks?

Here is a question beginners often miss. The DISCOM takes about two to four weeks to reply. An AI agent is a program that runs for seconds or minutes. It cannot sit and wait for weeks, and paying for a server that sits idle for weeks makes no sense.

The answer is: the agent does not wait. It **saves its notes, sets an alarm, and goes to sleep**. When the alarm rings, or when a human or a reply arrives, it wakes up, reads its notes and carries on. This is how every long-running agent in real life works.



IDEA	IN PLAIN WORDS	AWS TOOL WE WOULD USE
The notebook	Everything the agent knows about the project is stored, so nothing is lost between wake-ups.	DynamoDB (project record) and S3 (uploaded files, generated documents)
The alarm clock	A one-time reminder: 'wake me on day 21 and check whether the DISCOM replied'.	EventBridge Scheduler
The patient workflow	A step that pauses until a person clicks approve or an event arrives, then continues.	Step Functions with a wait-for-callback step (task token)
The messenger	Sends the committee a summary, and sends approved messages.	Amazon SES (email)

For the demo we will fake time

Judges cannot wait three weeks. The demo includes a 'time machine' button that moves the project clock forward (for example to day 25) and triggers the same wake-up logic. We must label this clearly as a simulation. The code path is the real one; only the clock is faked.

17 Keeping a human in charge

Good agents know when to stop and ask. Strands has a built-in feature for this called **interrupts**: the agent pauses before using a tool, hands the question to a person, and resumes with their answer (Sources 11). In SolarSabha, these are the stop signs:

THE AGENT WANTS TO...	RULE
Finalise a size recommendation	Show the assumptions and ask the committee to confirm.
Send any message to a vendor or official	Always pause. A named committee member must approve.
State that a rule applies	Must show the rule's text, source and the date we saved it.
Say a vendor is approved	Only if found in the loaded list. Otherwise say 'not verified'.
Do anything involving money or signatures	Not possible. No such tool exists.

18 Making an agent trustworthy

Anyone can build an agent that looks good in one demo. What judges and real users care about is whether it is right when it matters. Four habits make the difference:

HABIT	WHAT IT MEANS	HOW WE DO IT
Ground it in sources	The agent must not rely on memory for rules and numbers. It should look them up.	A small 'rulebook' file (subsidy rates, caps, steps) with source links and the date we checked. The agent must cite from it.
Structured answers	Ask the AI for answers in a fixed shape (like a form), so code can check them.	Bills and quotes are read into fixed fields. Missing fields are marked missing, never guessed.
Test with known answers	Make a few cases where you already know the right answer and check the agent against them.	10 sample bills and 6 sample quotes with known correct values. Re-run after every change.
Leave a trail	Record what the agent did and why, so a person can audit it.	Every tool call, input and output is logged to CloudWatch, and a summary saved to the project record.

Common ways agents go wrong (and the fix)

WHAT GOES WRONG	EXAMPLE	FIX
Invents a fact	Says a vendor is 'MNRE approved' with no proof.	Only allow claims backed by a tool result. Say 'not verified' otherwise.
Bad arithmetic	Miscalculates payback.	Maths lives in code tools.
Wanders off	Spends ten rounds re-reading the same bill.	Limit the number of rounds. Ask a human when stuck.
Over-trusts documents	A vendor PDF contains hidden text telling the AI to ignore its rules.	Treat document text as data, never as instructions. Keep rules in the system prompt.
Acts without asking	Sends an email by itself.	No send tool. Draft only, with a human approval step.

Safety point to mention
The 'hidden text in a document' risk is real for any agent that reads files from outsiders, and it is a good talking point with judges. Our defence is simple: documents are data, and the agent has no tool that could do harm.

PART D

TECH STACK

Every tool we plan to use, grouped by the job it does.

FRONTEND (WHAT THE COMMITTEE SEES)

React | Vite | SVG + CSS animation | AWS Amplify Hosting

BACKEND (THE ENGINE ROOM)

Python 3.12 | AWS Lambda | API Gateway (HTTP API) | AWS SAM

AI AND AGENTS (THE BRAINS)

Amazon Bedrock | A multimodal model on Bedrock | Strands Agents SDK | MCP (SolarRules server) | Pydantic structured output

LONG-RUNNING WORK (THE MEMORY AND ALARM CLOCK)

AWS Step Functions | EventBridge Scheduler | DynamoDB | Amazon S3

MESSAGES AND DOCUMENTS

Amazon SES | reportlab / python-docx | Amazon CloudWatch

DATA WE LOAD OURSELVES

Rulebook JSON (PM Surya Ghar) | CEA grid factor 0.710 | Sample approved-vendor list | Sample bills and quotes

TOOL	WHAT IT DOES FOR US, IN PLAIN WORDS
React + Amplify	The website the secretary uses: upload files, see the plan, approve drafts, view the timeline.
Lambda + API Gateway	Small pieces of backend code that run only when needed, so nothing is paid for while idle.
AWS SAM	Lets us describe the whole backend in one file and deploy it with one command.
Bedrock	Gives us access to the AI model (the brain) inside AWS.
Strands Agents SDK	AWS's open-source toolkit that runs the agent loop, handles tools, and supports human approvals and multi-agent teams.
MCP (SolarRules server)	A standard plug for tools. Our own small server exposes the rulebook and calculators so any agent can use the same trusted rules.
Multimodal reading	The Reader sends PDFs and photos straight to a Bedrock model that can see them. If another reader (such as Textract) works better on sample bills, we can swap it in.
Step Functions	Runs the long process and can pause until a human or an event continues it.

TOOL	WHAT IT DOES FOR US, IN PLAIN WORDS
EventBridge Scheduler	The alarm clock: wakes the agent on day 21 to check the DISCOM.
DynamoDB + S3	Notebook and filing cabinet: project state in the first, uploaded and generated files in the second.
SES	Sends the committee their summaries. New accounts start in a restricted 'sandbox' mode where mail goes only to verified addresses, which is fine for a demo.
CloudWatch	Records what the agent did, for debugging and for the demo's 'show your work' moment.

The hackathon accepts two ways to qualify: build locally with AWS open-source tools (for us, Strands Agents SDK and SAM) or deploy on AWS services. We plan to do both: develop locally with Strands and SAM, then deploy so the video can show AWS running it. Before 8 Oct, check in the Bedrock console which models (including those that read images) are available to our account in the Mumbai region.

PART E | THE PLAN

19 What we build in four days

The judges say one feature that runs beats five that almost do. So our promise is small and sharp: **give SolarSabha a society's bills and quotes, and it returns a trustworthy plan, warns about the risky quote, drafts the meeting papers, and handles a stalled DISCOM.**

MUST HAVE	SHOULD HAVE	LATER (not this week)
Reader on real PDFs and photos	Fairness table (equal vs by area)	Terrace photo scout
Plan with assumptions and CEA CO2	AGM notice and resolution drafts	Voice and local-language audio
Quote audit with live trace panel	Animated 2D arena replay	WhatsApp-chat reader
Hidden-instruction trap caught	Time machine, escalation draft	Real portal filing, payments
Secretary calling specialists	DISCOM reply gives Plan v1 to v2	Multiple states, 3D scene
	SolarRules MCP wired in	

DAY	WHAT WE BUILD	DONE WHEN...
Thu 8 Oct **See**	Get the live Reader working on the sample PDFs and photos. Deploy the SAM stack and the UI. Run the smoke test.	smoke_live passes; the Reader gets the sample bills and quotes right; the API and UI are online.
Fri 9 Oct **Judge**	Reader output feeds the same tools. Auditor explains flags. Secretary calls specialists. Trace panel shows live tool calls.	Verdicts match the expected GREEN, RED, AMBER. The hidden instruction is caught.
Sat 10 Oct **Follow**	Time machine wakes the Chaser. Escalation draft. DISCOM reply changes the plan. Arena replays real drafts. MCP wired.	A human-approved escalation and a Plan v2 appear on screen.
Sun 11 Oct **Ship**	Fixes only until noon. Record the video. README, blog, submit.	Submitted 3 hours early; every link opens in a private window.

Where we start
A starter repository already exists with the rulebook, all the plain-code tools, 19 passing tests, the offline pipeline, the local API, the React UI, the MCP server, sample data and the agent code. The agents are **written but not yet run against a live Bedrock model**. Day 1 is about making them work, not starting from zero. See Section 24.

Roles and before 8 October

ROLE	OWNS	FIRST JOB
A Frontend	frontend/: wire LIVE mode, polish the arena, trace panel, record the screen	Run make dev-ui and read every component.
B Backend and AWS	infra/, deploy, DynamoDB, scheduler wake, S3 upload	Run cfn-lint, then sam build when credentials are ready.
C Agents and AI	backend/solarsabha/agents/, the Reader's accuracy, the MCP server	Run scripts/smoke_live.py and fix what breaks.
D Story and quality	Rulebook verification, one real interview, evals, README, blog, video	Work through docs/RULEBOOK_VERIFICATION.md.

With two or three people, merge A with D and B with C.

Before 8 October

- The event says project work starts when the clock does, so until then **only learn and set up**. Ask on Discord whether preparing sample documents beforehand is allowed.
- Learn: the Strands quickstart and interrupts page; one Step Functions callback tutorial; how to enable a Bedrock model in the console.
- Everyone verifies their student profile on AWS Builder Center. Create the AWS account, set a budget alert, and check model access (including image and document input).
- Read the repository README and run the offline demo on your laptop.
- Have one real conversation with a society secretary or committee member. Ask what stopped them. Quote them in the video (with permission).

20 The three-minute video

There is no live demo, so this script is what judges see. All sample data is labelled on screen.

TIME	ON SCREEN	WHAT WE SAY
0:00-0:25	A real quote from a secretary or committee member; a terrace	'Solar can cut a society's lift and pump bill, and the government pays part. Most projects die from the process, not the maths.'
0:25-0:55	Upload 12 bills; table with an unclear value flagged	'The Reader agent turns bills into numbers and tells us what it is unsure about.'
0:55-1:20	Plan card and fairness tabs	'Size comes from the society's own use, not the subsidy ceiling. The maths runs in code. Every assumption is visible.'
1:20-1:55	Quote audit with the live trace; Vendor B's hidden instruction ignored	'One quote hides a command to the AI. It is ignored and flagged. The cheapest vendor is not on the list.'
1:55-2:25	Arena replay labelled SIMULATION; red vendor gets a proof request	'The agent drafts counters and the committee approves. Real vendors reply by email over days, so this replay is simulated.'
2:25-2:50	Time machine to day 50; escalation draft; DISCOM reply; Plan v1 to v2	'The Chaser notices the delay, drafts an escalation, and re-plans when the DISCOM limits capacity.'
2:50-3:00	Architecture slide; accuracy score	'Bedrock, Strands, MCP, Lambda, DynamoDB, EventBridge, S3. Humans approve everything.'

21 An honest expert review

Read this first
This review was written by the assistant, taking the view of an experienced AI engineer and a hackathon judge, and scoring against the criteria on the event page. It is not an official judging opinion. Use it to find weaknesses early.

CRITERION	BEFORE CUTS	AFTER CUTS	WHY
Real problem	8	8	Documented delays and traps; some evidence is from vendor guides.
Originality	7	7	Rooftop solar is a listed hint, so others will pick it. A process-manager agent is the difference.
Agentic merit	6	7	Genuine in follow-up and re-planning. Planner is code, not an agent.
Feasibility in 4 days	4	7	Seven agents, 3D, voice and MCP everywhere was too much. The starter repo helps.
Demo potential	8	8	Trap caught, live trace, time jump, plan changes.
AWS use	8	8	Meaningful, not decorative.
Trust and safety	8	9	Draft-only, cited rules, injection handled.
Measurable impact	6	6	kW and CO2 are computed from a sample society; outcomes are not measured.

Overall: about 6.5 out of 10 as first scoped; about 7.5 to 8 with the cuts above, if the live agents work.

The main criticisms and what we changed

CRITICISM	WHAT WE DID
Seven agents with 'roles' is architecture theatre. Maths is not an agent's job.	Coordinator plus three specialists. Planner is plain code.
Negotiating against simulated vendors proves little; an AI vendor will fold.	Reframed: the agent audits real quotes and drafts counters; the arena is a labelled replay. No savings claims.
3D costs a day you do not have.	2.5D animated SVG scene. 3D is a later idea.
MCP everywhere is a gimmick.	One real MCP server (SolarRules). Connectors only if time allows.
Your rules rest on weak sources.	A dated rulebook with a status for every rule and a verification checklist. 'Not verified' is shown openly.
No real user yet.	One real interview before the video, quoted with permission.

Questions a judge may ask (prepare answers)

- **Isn't this a calculator with a chatbot?** The calculator is a small part. The value is auditing real quotes, running a group decision and following up for weeks.
- **Who are the vendors?** Simulated in the demo and labelled. The reading and auditing work on real PDFs.
- **How do you know the rules are right?** A dated rulebook with sources. Anything not from an official document is marked 'verify'.
- **What if the AI is wrong?** Numbers come from code, nothing can be sent, and 'not verified' is an allowed answer.
- **Why not a vendor's free report?** The vendor is selling. This works for the society.

22 What could go wrong

RISK	WHAT WE DO ABOUT IT
Live agents have never run against Bedrock in this project	Day 1 starts with the smoke test. Offline mode runs the full loop meanwhile, so the demo never depends on one untested call.
Rules differ by state and change	Model one state. Keep rules in a dated rulebook with links. The agent cites it and tells the committee to confirm with the DISCOM.
Real vendor and DISCOM data is not available	Use clearly labelled sample data. Say so in the video.
The AI misreads a bill	Show what was read next to the file, mark uncertain fields, let a person correct them. Report the eval score honestly.
Judges say 'this is just a calculator'	Lead with the quote audit, the trace panel and the follow-up loop.
The agent claims something false	Claims need tool evidence. 'Not verified' is allowed.
Too much for four days	Cut from the bottom of 'Should have'. Never cut Reader, quote audit or the time-jump loop.
The UI was built but not checked visually in a browser	Run it on Day 1 and fix layout problems first.

23 Questions for the team to decide together

- Which state and DISCOM do we model, and who can sanity-check the process for us?
- Is the society the only user, or do we also support a single flat owner?
- Which Bedrock model do we use, and does it read both photos and PDFs well?
- Who owns which role (Section 19), and who talks to a real committee member?
- Which track are we in? SolarSabha is Track 03 (Waste and Energy). Paani Aaya is Track 02. Each track has its own winner, so two strong projects in different tracks do not compete with each other.

24 The repository, in plain words

Everything the team needs is in one folder called **solarsabha**. The README at its top explains how to run it. This page is the map.

```
solarsabha/
  README.md  Makefile  .env.example
  docs/        guide (this PDF), ARCHITECTURE, AGENTS, TEAM_PLAN,
               DEMO_SCRIPT, JUDGE_QA, RULEBOOK_VERIFICATION, SETUP_AWS
  backend/
    solarsabha/
      data/rulebook.json        dated, cited rules (the single source of truth)
      tools/                    plain code: sizing, quotes, fairness, timeline,
                                negotiation, drafts
      pipeline.py               the whole loop, no AWS and no LLM needed
      agents/                   Strands agents, prompts, multimodal Reader, MCP link
      handlers/                 API Lambda, scheduled wake-up Lambda
    mcp_server/                 SolarRules MCP server
    samples/                    sample society, bills, quotes (with the trap)
    tests/                      19 tests
  frontend/                     React app: plan, audit, arena, trace, timeline, drafts
  infra/template.yaml           AWS SAM: DynamoDB, S3, HTTP API, Lambdas
  scripts/                      demo_offline, run_evals, smoke_live, dev_server ...
```

First five minutes for each teammate

```
make setup      # install dependencies
make test       # 19 tests, no AWS needed
make demo       # run the whole loop in the terminal
make dev-ui     # open the app on bundled demo data
```

What is real and what is not

PART	STATUS
Rulebook, calculators, quote audit, timeline, negotiation logic, drafts	Written and unit-tested.
Offline pipeline, local API, MCP server	Written and tested.
React app	Builds. Not yet checked visually in a browser.
Strands agents and the multimodal Reader	Written for strands-agents 1.58. Construct and register tools in tests. **Never run against live Bedrock.**
AWS SAM template	Passes cfn-lint. **Never deployed.**
Vendors, DISCOM replies, the clock, the approved-vendor list	Simulated and labelled.
Rules marked 'verify'	Must be checked against official guidelines before the demo.

25 Words you will hear, explained

WORD	MEANING
Agent	An AI that decides what to do next, uses tools, checks results, and repeats until the goal is done or a human is needed.
LLM (language model)	The AI that reads and writes text. The brain inside an agent.
Tool	A function the agent can call to do something real: read a file, calculate, look up, draft.
System prompt	The standing instructions given to the agent before any conversation.
Agent loop	Think, act, look at the result, repeat.
Human in the loop	A person approves or corrects at key points.
Multi-agent	Several specialist agents coordinated by one.
MCP	Model Context Protocol: a standard plug that lets any agent use an outside tool.
Multimodal	An AI that can read more than text, such as photos and PDFs.
Prompt injection	Hidden text in a document that tries to give the AI new orders. We treat documents as data and ignore such text.
Structured output	Forcing the AI to answer in a fixed form, like filling a form, so code can check it.
Guardrails	Hard limits that hold even if the AI makes a mistake.
Serverless	Code that runs only when needed, with no machine for us to look after.
DISCOM	Distribution company: the local electricity company.
RWA / GHS	Resident Welfare Association / Group Housing Society.
kW and kWh	kW is the size of the system. kWh (a 'unit' on your bill) is the amount of energy used or made.
Net metering	A meter that records electricity sent to the grid as well as taken from it, so the bill is reduced.
Virtual net metering	One shared solar system whose bill credits are split across many meters.
Empanelled vendor	An installer on the official approved list.
Commissioning	The official sign-off that the system is installed and working.
DBT	Direct Benefit Transfer: the subsidy paid into a bank account.
CEA grid factor	The official figure for how much CO2 each unit of grid electricity causes.

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10. Heaven Green Energy. PM Suryaghar for society flat owners, common area rules 2026 (AGM steps and stage timings); Reslink EPC guide 2026.

11. Strands Agents documentation: strandsagents.com (agent loop, tools, interrupts, multi-agent patterns).

12. WeMakeDevs x AWS. Environmental Hacks event page and rules: wemakedevs.org/aws/env

13. Model Context Protocol and Strands MCP documentation: strandsagents.com (MCP tools).

Note: sources 3, 6 to 10 are company or think-tank material, not government rules. Treat their timelines and 'traps' as warnings to verify against the official scheme guidelines.